

Ex.no:3	DATA CLEANING

Aim:

Algorithm:

Check Missing Values:

1. Start.
2. Import the pandas library.
3. Load the dataset into a DataFrame (df).
4. Display dataset information using df.info().
5. Check missing values using df.isnull().
6. Count missing values using df.isnull().sum().
7. Check missing values using df.isna().
8. Check non-missing values using df.notnull().
9. Display the results.

Drop Missing Values

1. Start.
2. Import the pandas library.
3. Load the dataset into a DataFrame (df).
4. Drop missing rows using dropna().
5. Drop rows (axis=0) or columns (axis=1).
6. Use how='any' to remove rows/columns containing any missing value.
7. Use how='all' to remove rows/columns only if all values are missing.
8. Use thresh=N to keep rows/columns with at least N non-missing values.
9. Use subset=['column_name'] to drop rows based on specific columns.
10. Use inplace=True to modify the original DataFrame.
11. Display the cleaned DataFrame.

Program:

```
import pandas as pd

# Load dataset
df = pd.read_csv("hotel_bookings.csv")

print("===== DATASET INFO =====")
df.info()

print("\n===== MISSING VALUES IN EACH COLUMN =====")
print(df.isnull().sum())

print("\n===== TOTAL MISSING VALUES =====")
print(df.isnull().sum().sum())

print("\n===== ANY MISSING VALUES? =====")
print(df.isnull().values.any())

print("\n===== NON-MISSING VALUES COUNT =====")
print(df.notnull().sum())

import pandas as pd

df = pd.read_csv("hotel_bookings.csv")

# Fill all missing values with 0
df_constant = df.fillna(0)
print("===== CONSTANT VALUE =====")
print(df_constant.head())

# Fill 'children' column with mean
```

```
df_mean = df.copy()
df_mean["children"] = df_mean["children"].fillna(df_mean["children"].mean())
print("\n===== MEAN =====")
print(df_mean.head())

# Fill 'children' column with median
df_median = df.copy()
df_median["children"] = df_median["children"].fillna(df_median["children"].median())
print("\n===== MEDIAN =====")
print(df_median.head())

# Fill 'country' column with mode
df_mode = df.copy()
df_mode["country"] = df_mode["country"].fillna(df_mode["country"].mode()[0])
print("\n===== MODE =====")
print(df_mode.head())

# Forward Fill
df_ffill = df.ffill()
print("\n===== FORWARD FILL =====")
print(df_ffill.head())

# Backward Fill
df_bfill = df.bfill()
print("\n===== BACKWARD FILL =====")
print(df_bfill.head())

import pandas as pd

df = pd.read_csv("hotel_bookings.csv")

# Drop rows with missing values
df_drop_rows = df.dropna()
```

```
print("===== DROP ROWS =====")
```

```
print(df_drop_rows.head())
```

```
# Drop columns with missing values
```

```
df_drop_cols = df.dropna(axis=1)
```

```
print("\n===== DROP COLUMNS =====")
```

```
print(df_drop_cols.head())
```

```
# Drop rows having any missing value
```

```
df_any = df.dropna(how="any")
```

```
print("\n===== DROP HOW='ANY' =====")
```

```
print(df_any.head())
```

```
# Drop rows where all values are missing
```

```
df_all = df.dropna(how="all")
```

```
print("\n===== DROP HOW='ALL' =====")
```

```
print(df_all.head())
```

```
# Keep rows with at least 30 non-null values (32 columns total)
```

```
df_thresh = df.dropna(thresh=30)
```

```
print("\n===== THRESH=30 =====")
```

```
print(df_thresh.head())
```

```
# Drop rows where 'country' or 'children' is missing
```

```
df_subset = df.dropna(subset=["country", "children"])
```

```
print("\n===== SUBSET (country, children) =====")
```

```
print(df_subset.head())
```

```
# Drop missing values permanently
```

```
df.dropna(inplace=True)
```

```
print("\n===== AFTER inplace=True =====")
```

```
print(df.head())
```

Output:

```
===== MISSING VALUES IN EACH COLUMN =====
hotel                                0
is_canceled                          0
lead_time                            0
arrival_date_year                     0
arrival_date_month                    0
arrival_date_week_number              0
arrival_date_day_of_month             0
stays_in_weekend_nights               0
stays_in_week_nights                 0
adults                               1
children                             1
babies                               1
meal                                  1
country                              472
market_segment                       1
distribution_channel                  1
is_repeated_guest                    2
previous_cancellations                2
previous_bookings_not_canceled        2
reserved_room_type                    2
assigned_room_type                    2
booking_changes                       2
deposit_type                          2
agent                                13520
company                              91473
days_in_waiting_list                 2
customer_type                         2
adr                                   3
required_car_parking_spaces           3
total_of_special_requests             3
reservation_status                    3
reservation_status_date               4
dtype: int64

===== TOTAL MISSING VALUES =====
105505

===== ANY MISSING VALUES? =====
True
```

===== CONSTANT VALUE =====

	hotel	is_canceled	lead_time	arrival_date_year	arrival_date_month	arrival_date_week_number	arrival_date
0	Resort Hotel	0	342	2015	July	27	
1	Resort Hotel	0	737	2015	July	27	
2	Resort Hotel	0	7	2015	July	27	
3	Resort Hotel	0	13	2015	July	27	
4	Resort Hotel	0	14	2015	July	27	

5 rows x 32 columns

===== MEAN =====

	hotel	is_canceled	lead_time	arrival_date_year	arrival_date_month	arrival_date_week_number	arrival_date
0	Resort Hotel	0	342	2015	July	27	
1	Resort Hotel	0	737	2015	July	27	
2	Resort Hotel	0	7	2015	July	27	
3	Resort Hotel	0	13	2015	July	27	
4	Resort Hotel	0	14	2015	July	27	

5 rows x 32 columns

===== MODE =====

	hotel	is_canceled	lead_time	arrival_date_year	arrival_date_month	arrival_date_week_number	arrival_date
0	Resort Hotel	0	342	2015	July	27	
1	Resort Hotel	0	737	2015	July	27	
2	Resort Hotel	0	7	2015	July	27	
3	Resort Hotel	0	13	2015	July	27	
4	Resort Hotel	0	14	2015	July	27	

5 rows x 32 columns

===== FORWARD FILL =====

	hotel	is_canceled	lead_time	arrival_date_year	arrival_date_month	arrival_date_week_number	arrival_date
0	Resort Hotel	0	342	2015	July	27	
1	Resort Hotel	0	737	2015	July	27	
2	Resort Hotel	0	7	2015	July	27	
3	Resort Hotel	0	13	2015	July	27	
4	Resort Hotel	0	14	2015	July	27	

Result: